VORTEXA 3.0

12 HRS HACKATHON

POWERED BY A.I.R.S

DOMAIN & NO : 3, B-03

PROBLEM STATEMENT TITLE : MILESTONEVAULT: ADVERSARIAL SMART-CONTRACT ESCROW

TEAM NAME : TORVALDS

TEAM ID : XQR1520246

TEAM LEADER : Omkar Rane

TEAM MEMBERS : Omkar Rane, Pranay Patil

SEALSTONE

Escrow that can't be tricked

PROBLEM STATEMENT

Milestone escrow breaks in three ways, *and most projects fix only one.*

1 Ambiguous agreements

Agreements are written in prose. Vague terms hide disputes before any money moves.

"to the mentor's satisfaction"

"paid from the remaining funds"

2 Swapped deliverables

Approval isn't tied to a specific file, so what was reviewed can differ from what gets paid.



reviewed ≠ delivered

3 Buggy money code

A double release, a missing check or a re-entrant call can drain funds that can't be recovered.

3.000 ETH

taken from a naive escrow in one attack

THE BRIEF SAYS

AI may propose a checklist, but a person must confirm it, ambiguities must be resolved, and AI output must never be written to the contract directly.

IDEA TITLE / PROPOSED SOLUTION

Sealstone: three layers of integrity, *so nobody can trick the payment.*

1 AGREEMENT INTEGRITY

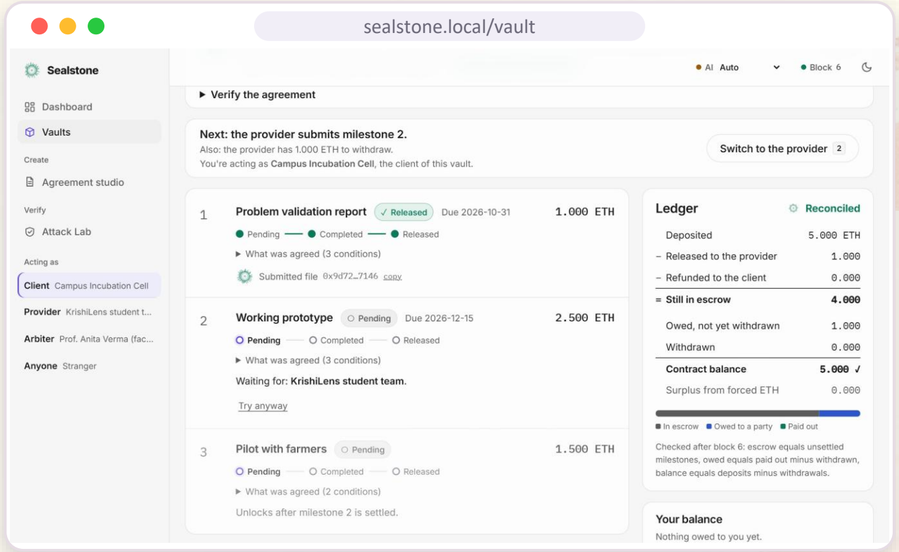
AI proposes and flags unclear terms. Rules re-check, a person resolves every flag, and both sides sign the hash.

2 DELIVERABLE INTEGRITY

The provider commits the file's SHA-256. The client can approve only that exact file.

3 FUND INTEGRITY

Role-gated states, pull payments and a re-entrancy guard. A live ledger reconciles after every action.



Every milestone, and a ledger that reconciles

AI informs. Rules decide.

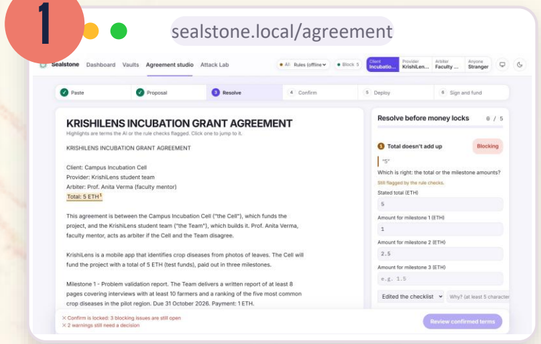
Every money-moving decision is made in Solidity. AI output never reaches the contract.

PRODUCT IN ACTION

A working product, not a mock-up: *judges can try every step live.*

1

sealstone.local/agreement



AI proposes and flags

Unclear terms are highlighted. Confirm stays locked until each is resolved.

2

sealstone.local/dashboard

DASHBOARD SCREENSHOT

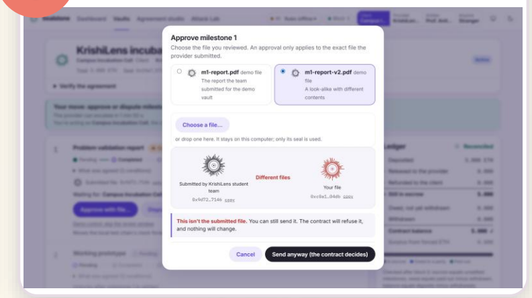
Placeholder: the final dashboard capture goes here

One dashboard, every vault

What's locked, owed and waiting, with the ledger reconciled.

3

sealstone.local/vault

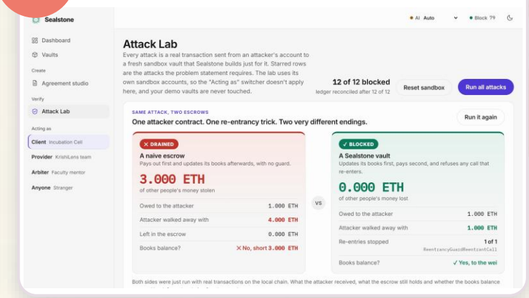


A swapped file is refused

Two different seals. The contract reverts and funds don't move.

4

sealstone.local/attack-lab



12 real attacks, one click

Each is blocked, and the ledger reconciles after every one.

LIVE DEMO PATH *about 5 minutes*

1. **Extract**
2. **Resolve flags**
3. **Confirm + sign**
4. **Submit file**
5. **Swap it: refused**
6. **Run 12 attacks**

TECHNICAL APPROACH 1/3

From a paragraph to a protected payment: *AI proposes, rules decide, people confirm.*

OFF-CHAIN *AI proposes, rules and people decide*

1 Agreement text
pasted prose, any format

→

2 AI proposes
milestones and ambiguities (Ollama, Gemini or rules)

→

3 Rules re-check
wei arithmetic, parties, quotes found in the text

→

4 A person resolves
edit, remove or acknowledge every flag

→

5 Confirmed checklist
the only version that can be deployed

↓

AI BOUNDARY: only the confirmed, hashed version crosses

ON-CHAIN *Solidity decides*

6 Seal on-chain
keccak256 hash stored, both parties sign

←

7 Fund exactly
the client deposits the exact total

←

8 Deliver
the provider commits the file's SHA-256

←

9 Approve exact hash
or dispute: the arbiter decides

←

10 Pull and reconcile
withdraw, then the ledger checks itself

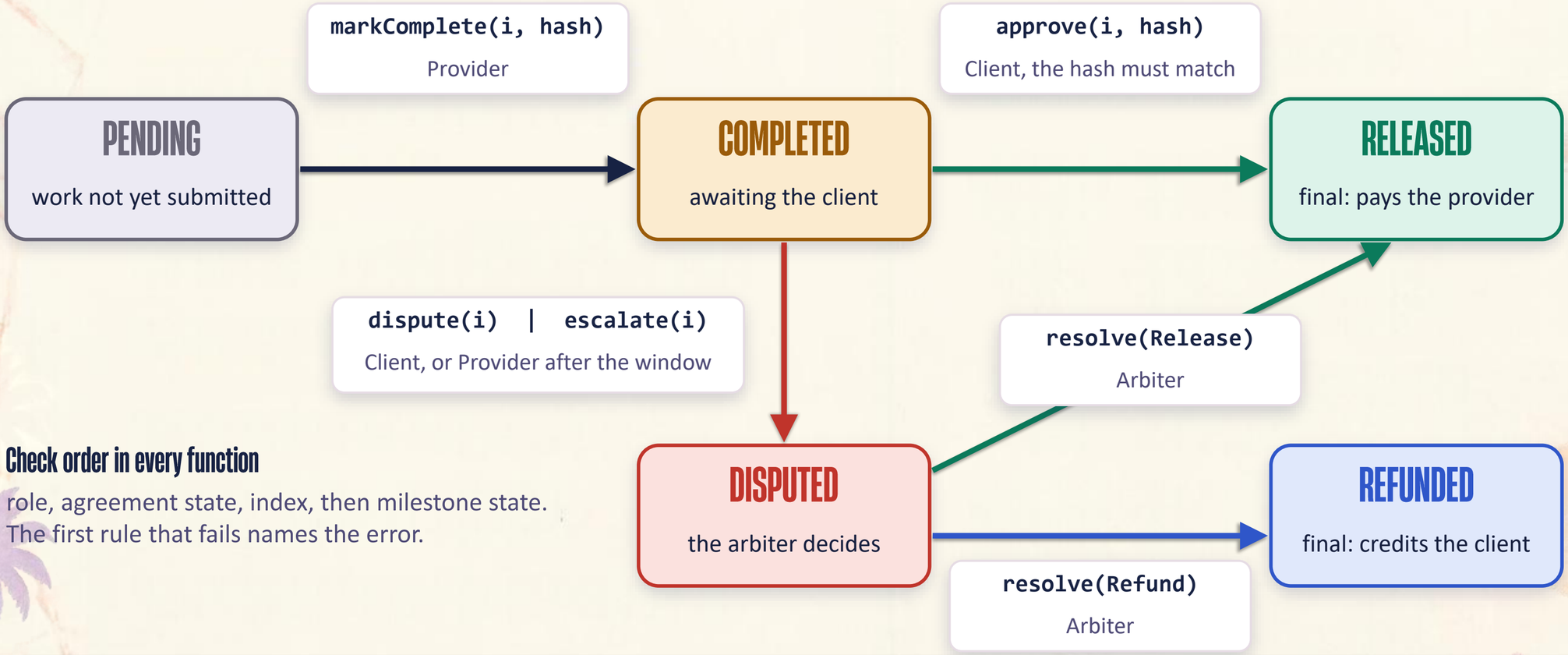
COLOUR KEY

AI-proposed | Rules | Person | On-chain

Every refusal is a named error, and a refused call changes nothing.

TECHNICAL APPROACH 2/3

One state machine per milestone. *Anything not on the map reverts with a named error.*



Check order in every function

role, agreement state, index, then milestone state.
The first rule that fails names the error.

Final states never change.

A second release, an unauthorized refund or a skipped step reverts with a named error.

TECHNICAL APPROACH 3/3

Three small parts, *and the AI service can never touch the chain.*

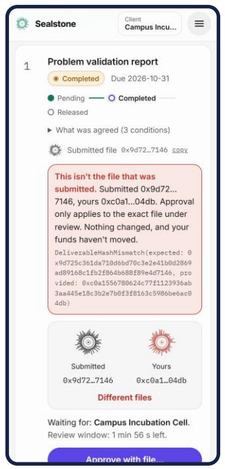
WEB APP

Agreement studio

Vault dashboard

Attack Lab

React, TypeScript, ethers v6



HTTP

AI SERVICE

extract, validate, gate

Ollama, Gemini or built-in rules

never connects

LOCAL CHAIN

MilestoneVault.sol

OpenZeppelin guard

Attack contracts, test only

Solidity 0.8.24, Hardhat

SHARED PACKAGE

One implementation of hashing, validator and gate

JSON-RPC: sign, fund, approve, withdraw

Local fallback.

With Wi-Fi off it still works: rules extractor, local chain, bundled fonts. No paid API, GPU or cloud.

FEASIBILITY AND VIABILITY

Built, tested and reproducible in 12 hours, *with no paid service.*

BOOKLET RULE	EVIDENCE
✓ Working MVP, testable live	npm run dev: studio, vault, Attack Lab
✓ Reproducible live testing	demo:reset, a seeded vault, 383 tests
✓ AI used meaningfully	extracts milestones, flags ambiguity
✓ Checkable rules decide	validator and Solidity; AI never overrides
✓ Fictional data only	invented agreements, test ETH, local chain
✓ Local fallback	rules extractor and local chain, Wi-Fi off

138
contract tests passing

100%
line coverage of the escrow

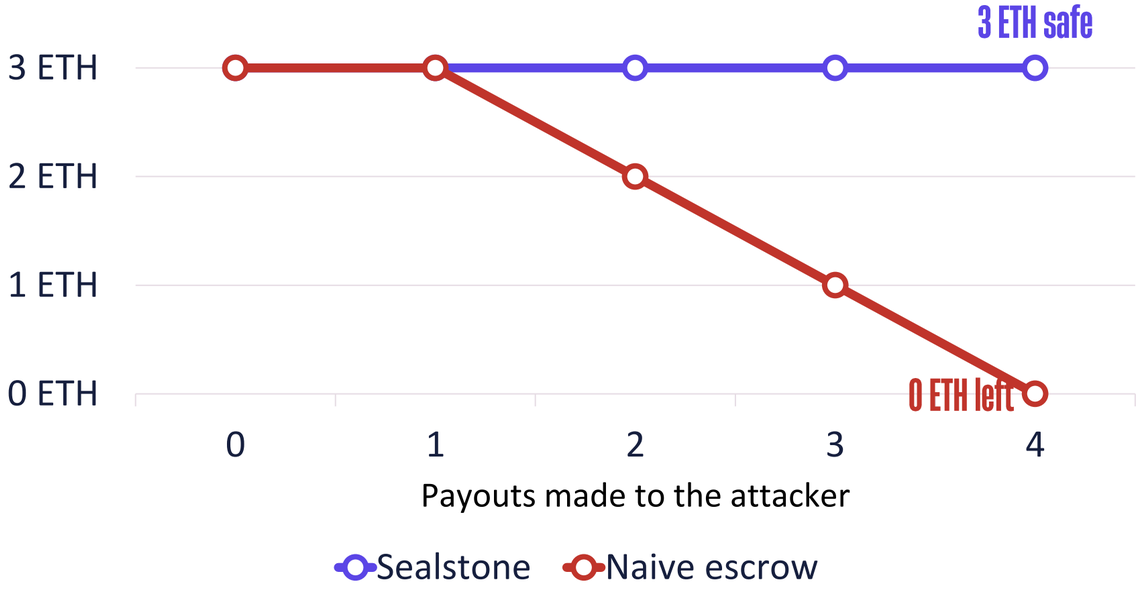
1,000
fuzz steps, 8 invariants

12/12
real attacks blocked

```
$ npm run dev
$ npm run test:contracts
$ npm run demo:reset
```

IMPACT AND BENEFITS

Fewer disputes, no swapped files, *and no hidden intermediary to trust.*



One re-entrancy attack, payout by payout. In the Attack Lab's contrast run the attacker takes 4 ETH and 3 ETH of other users' money is lost.

1 Freelancers and clients

Pay only for the exact file you approved. Disputes go to a named referee.

2 Grant programmes

One dashboard shows what's locked, owed and waiting across every agreement.

3 Auditors and finance

The ledger reconciles after every action, and the rules are tests anyone can run.

WHAT'S NEXT

- Factory for many agreements
- A committee as arbiter
- Pilot on a public testnet

DIFFERENTIATOR

Most escrow demos secure the money. *Sealstone also secures the agreement and the file.*

AGREEMENT INTEGRITY

Vague terms can't reach the contract

"to the mentor's satisfaction"

↓

A live demo, set by a person

LIVE PROOF

`agreement studio, Extract`

DELIVERABLE INTEGRITY

Approval is bound to the exact file



`DeliverableHashMismatch`

LIVE PROOF

`rejects approval of a swapped deliverable`

FUND INTEGRITY

Forced-in money can't break the books

0.300 ETH

surplus, never withdrawable

✓ books still reconcile

LIVE PROOF

`attack: Force ETH in with selfdestruct`

ADVERSARIAL PROOF

You fire the attacks yourself

12 / 12

real transactions, all blocked

Run all attacks

LIVE PROOF

`Attack Lab, Run all attacks`

Don't take our word for it. Pick a role in the demo vault, or press Run all attacks.

AI never moves money

Works with Wi-Fi off

Honest limits: one contract per agreement, a trusted arbiter, and a hash proves identity, not quality.

SEALSTONE *escrow that can't be tricked*

Team TORVALDS · XQR1520246 · Omkar Rane (leader), Pranay Patil

THANK YOU!

Game's Still On

VORTEXA 3.0

AIRS

